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ABSTRACT

The environmental pollution caused by organic molecules as Tetracyclines and Oxytetracyclines presents a foremost concern, as high levels of these antibiotics when released into the ecosystem are accumulated and are producing overall contamination. Consequently, the need for selective, sensitive, fast, easy to handle, and low-cost early monitoring detection systems is growing. Electrochemical sensors based on modified carbon electrodes involve relevant materials for antibiotics' detection, which allows the early controlling of the spread of these antibiotics into the environment, thus, safeguarding human health against possible contamination. The present review paper has aimed to illustrate examples of newly developed electrochemical bio(chemo)sensors with their real application fields for the sensitive detection of Tetracycline as well as Oxytetracycline in various matrices. Future forecast for the utilisation of these sensors is also discussed.

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Tetracycline; oxytetracycline; electrochemical detection; carbon-based electrodes; chemo(bio)sensors

1. Introduction

Recently, the impulsive residues of antibiotics, especially Tetracyclines used for medical, industrial, or agricultural purposes are a source of environmental pollution due to their accumulation, stability, and persistence in aquatic (surface, drinking, ground, and waste-waters) as well as in terrestrial (soil and sediment) ecosystems [1] which makes their elimination a difficult task. After long exposure to excessive quantities, too harmful effects associated with dangerous hazards that threaten human health by toxic effects and bacterial resistance are arising [2], and manifesting in various diseases [3,4] such as gastrointestinal disturbances, renal dysfunctions, intracranial hypertension as well as skin infections, e.g. rosacea and perioral dermatitis. Moreover, exposure to Tetracycline antibiotics induces deficiencies in protein production due to their direct attachment to the ribosome, which blocks the formation of proteins in organisms [5]. The water sources'

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contamination by Tetracycline antibiotics comes mainly from their direct consumption by humans during medical treatments, which secretes them through urine discharges into urban wastewater, or indirectly via eating the meat of animals treated by Tetracyclines in the breeding stage [6]. Hence, many national and international organisations as the World Health Organization (WHO) and European Union (EU) [7] have set permissible thresholds regarding limit amounts of Tetracyclines in drinking water, food, and wastewater. Therefore, the development of appropriate analytical procedures for the detection and determination of Tetracycline residuals in water, soil, and food samples is of paramount importance.

Several conventional methods are used for the determination of such antibiotic residues [8] as microbiological methods [9,10], viz. immunoassays [11], enzyme immunoassays [12,13], and antibiogram tests. Chromatographic methods are also utilised to detect and quantify tetracyclines as Thin-Layer Chromatography (TLC) [14], High-Performance Liquid Chromatography (HPLC) [15–18], and Capillary Electrophoresis (CE) [19–22]. Spectroscopic techniques are employed as well [23] e.g. spectrofluorimetry [24], UV-visible spectroscopy [25], chemiluminescence (CL) [26] and colorimetry [27]. Combined analytical techniques were also employed, outstandingly, UV-visible spectroscopy coupled to HPLC [28], and to Diode Array Detectors (UV-DAD) [29], Pulsed Amperometric Detectors (PAD) [30,31], Fluorescence spectroscopy Detectors (FD) [32], and Mass spectrometry detectors (MD) [33,34], including Capillary Electrophoresis coupled to Mass spectrometry (CE-MS) [35,36].

The methods stated above showed their usefulness and potential for the Tetracyclines determination and quantification. However, they seem to be expensive, tedious [3], complicated and require mastery and maintenance [8]. Consequently, innovative, simple, and affordable alternatives are necessary.

Utilising electrochemical techniques [5] viz. Cyclic Voltammetry (CV) [37], Square Wave Voltammetry (SWV) [38], Differential Pulse Voltammetry (DPV) [39], Electrochemical Impedance Spectroscopy (EIS) [40] and ElectroChemiLuminescence (ECL) [41], huge progress has been achieved providing affordable, convenient, and accurate alternatives [3] for the detection of Tetracyclines in different samples. Moreover, an enormous focus on modified carbon electrodes-based electrochemical sensors has arisen, generally, the modifier choice plays a major role in the selection of these electrodes since it confers to them high adsorption properties and fast electron transfer kinetics, enhancing the sensitivity and improving the detection limits of these sensors towards Tetracyclines quantification.

Many works have focused on the detection of the Tetracyclines' family by using different biological modifiers as aptamers (DNA, enzyme ...), and chemical modifiers (oxides, clays, acids, or else metallic nanoparticles), following this prospect, the present review paper aims at highlighting the electrochemical detection of Tetracycline (TC) and Oxytetracycline (OTC) employing carbon electrodes-based bio and chemosensors modified via various biological or chemical materials (Figure 1).

2. Tetracycline (TC) and Oxytetracycline (OTC)

Tetracycline (TC) and Oxytetracycline (OTC) (Figure 2) are wide-spectrum antibacterial, anticancer, antiprotozoal, and antimalarial antibiotics [5,8,10]. Generally, they are obtained by fermentation of Streptomyces rimosus which is a soil actinomycete [42].

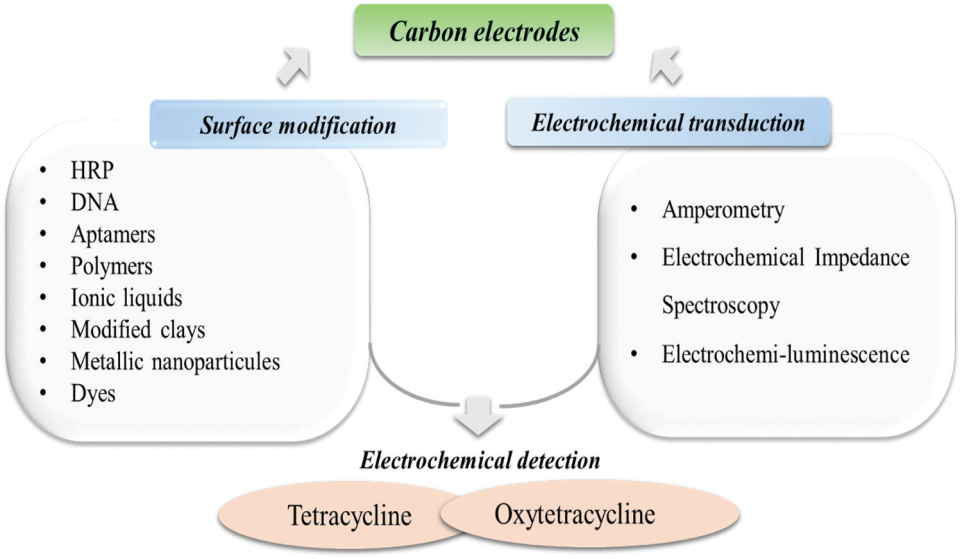


Figure 1. Chemical structures of (A) Tetracycline and (B) Oxytetracycline [10].

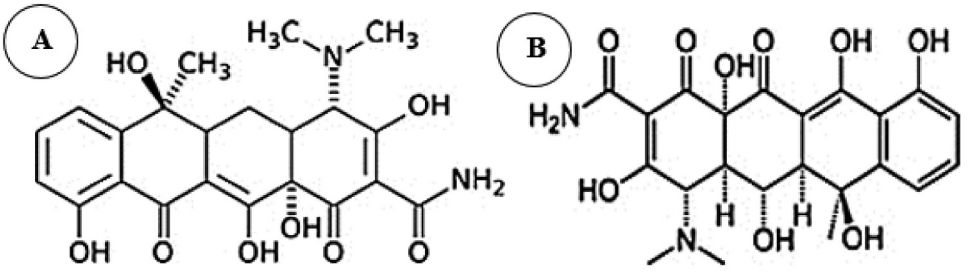


Figure 2. Chemical structures of (A) Tetracycline and (B) Oxytetracycline [10].

Since their discovery in 1940, these antibiotics have been practically and broadly adopted for humans and animals, notably for bacterial diseases' treatment such as gram-positive and gram-negative [3], for respiratory, urethritis, and skin infections treatments, as well as multi-resistant malaria treatment [2]. Owing to their low cost, TCs and OTCs are widely used in agriculture and aquaculture as nutritional additives for poultry, fish, sheep, and cattle [43].

3. Applied electrochemical transduction methods for TCs and OTCs detection based on modified carbon electrodes

The sensitivity of a detection platform depends essentially on the applied transduction method. Since the transduction mechanism provides amplification of the detection signal relative to the target analyte, a reinforcement of the detection limits and other sensing

characteristics is achieved [44]. Thus electrochemical methods as amperometry, cyclic (CV), linear (LSV), pulse (SWV, DPV ...), and adsorptive pulse voltammetry, as well as electrochemical impedance spectroscopy (EIS) and electrochemiluminescence (ECL) are the most utilised transduction techniques for TCs and OTCs detection.

3.1. Amperometric transduction

Amperometry is based on the measurement of current intensities resulting from a redox reaction of an electroactive analyte at a constant potential between reference and working electrodes [45–47]. This method requires a three electrodes' assembly (working, reference and counter-electrode), and has the advantage of being selective since the resulting current intensities are strictly proportional to the concentration of the electroactive analyte present in the electrolyte. Moreover, amperometric transduction provides information about the process and the electron transfer kinetics between the analytes and the electrode surface, which helps in understanding the detection mechanism.

Voltammetric techniques [37,39] are part of amperometry, these methods are highly desirable for electroanalysis, especially for current transduction resulting from electrochemical sensors designed for tetracyclines detection.

3.1.1. Cyclic Voltammetry (CV)

Cyclic voltammetry involves, at a constant rate, a forward and backward sweeping in a triangular waveform of applied potential to the working electrode in such a way to induce oxidation and subsequent reduction of the electroactive analyte present in solution or adsorbed on the electrode surface, which leads to a current flow between the working and the counter electrodes [48]. Consequently, the recording of this flow allows obtaining specific current-potential voltammograms to the studied system. During CV studies, the scan rate must be sufficiently long so that the chemical/electrochemical reactions can occur to a significant level [49], thus yielding information about redox potential, as well as diffusion kinetic parameters related to the electrochemical reactions within the system could be gotten.

3.1.2. Linear Sweep Voltammetry (LSV)

LSV is an electroanalytical technique where the working electrode potential varies linearly over the analysis time. This technique consists of sweeping a potential (Ei) where no reaction occurs towards a higher potential value relative to oxidation or more negative potential value relative to reduction of the studied electroanalyte. During this analysis, the measured current intensity generates the current-potential (I-E) curve named voltammogram with a typical shape indicating the analyte's red/ox characteristics as well as the main parameters of the involved phenomenon [50].

3.1.3. Pulse voltammetry

3.1.3.1. Square Wave Voltammetry (SWV). SWV is a very rapid pulse-based voltammetric technique that offers high rejection as well as excellent sensitivity of capacitive currents. It provides a reliable determination of the analyte with detection limits comparable to those obtained by chromatographic or spectroscopic techniques [51]. The SWV measurement involves the application of a different potential compared to the

electroactive element's redox potential to ensure the working electrode's polarisation [52], consequently producing a bell-shaped current-potential curve useful for deriving characteristics of the generated electrochemical reactions on the electrode surface.

3.1.3.2. Differential Pulse Voltammetry (DPV). The DPV consists of applying pulsed amplitudes to linear growth potential. Like SWV, the choice of the imposed potential to the electrode for performing the analysis must be strictly different from the faradic reactions' potentials [51]. Hence, these potentials increase progressively between impulsions, yielding an analytical signal related to the difference between the currents measured before and after these impulsions.

3.1.3.3. Adsorptive Pulse Voltammetry. Adsorption pulse voltammetric techniques (ads SWV or ads DPV) work exactly like normal SWV and DPV, nevertheless, these techniques are suitable for non-electroactive analytes, since they involve a spontaneous adsorption process providing the preconcentration step of the target analyte on the electrode surface. For the validity of these techniques, the analyte's concentration must be within the linear cross-section range of the adsorption isotherm [53]. Also, adsorption pulse voltammetric methods are appropriate for trace analysis since the analyte concentrations should be ranging between 10^{-9} to 10^{-7} M.

3.2. Transduction by Electrochemical Impedance Spectroscopy (EIS)

The EIS studies the sensitive phenomena occurring on the electrode surface as well as the changes in electrical properties at this surface during an electrochemical process. The EIS's operating principle is based on the control, in a given frequency range [54] of the interactions produced on the detection platform during a reaction between the electroactive analyte dissolved in the solution and the modifier located at the electrode surface, leading to equivalent electrical circuits of the involved phenomena on the electrode/solution interface.

3.3. ElectroChemiLuminescence (ECL) transduction

ECL is a linear, stable and selective technique that involves measuring the emitted light by an analyte called a co-reagent during an electrochemical reaction produced on an electrode surface [41]. ECL is based on the excitation of a luminophore by the co-reagent present in the sample solution, leading to the emission of photons at characteristic wavelengths during the return of the luminophore to the fundamental state. This results in an analytical signal proportional to the amount of co-reagent.

4. Carbon materials employed in carbon-based working electrodes

Due to their wide range of electroactivity, low cost, and low toxicity which reduces the risk of environmental pollution, carbon-based electrodes are excellent alternatives to metallic ones as gold, platinum, mercury, or bismuth electrodes. Typically, these electrodes are based on graphite, glassy carbon, diamond, carbon fibres, and nanotubes which are easily

functionalised and chemically unaffected, making the modification process easier along with reaching low detection limits when quantifying electroactive analytes.

4.1. Carbon Fibre (CF)

Carbon fibres are usually obtained by carbonisation followed by graphitisation of polyacrylonitrile (PAN) or carbon pitch. These fibres have very high tensile strengths ranging from 200 to 800 GPa which are appreciable mechanical properties during the electrochemical process. Carbon fibres are frequently used as embedded systems in polymer matrices [55].

4.2. Carbon Screen Printed Electrodes (C-SPEs)

C-SPEs are composed of working electrodes made up of conductive carbon-based inks, where these inks contain carbon blended with binder pastes, organic solvents, or some additives that supply functional characteristics [56].

4.3. Glassy Carbon (GC)

GC is an amorphous material made from precursors as phenolic resins and polycarbodiimide carbonised at over 1600°C. GC has a large potential window and an easily regenerable surface, which is highly recommended for electroanalytical tests [57].

4.4. Graphite Paste (GP)

GP electrodes are made up of four components: graphite, binder, electrode body (glass or teflon tube), and a conductive metallic wire for electrical contact. The employed graphite in the paste preparation must be of high purity, low adsorption capacity, and in the form of a micrometric particle size powder. The binder, which is a substance allowing the graphite to be compacted in the electrode body, is generally a liquid, non-electroactive, hydrophobic, low-volatility organic substance such as hydrocarbons viz. aliphatic, aromatic and halogenated hydrocarbons, mineral oils such as silicone or kerosene, as well as polymeric materials such as epoxy resins, styrene-acrylonitrile and polyester [58].

4.5. Carbon NanoTubes (CNTs)

CNTs are semiconductors made up of functionalised graphene layers wraps. CNTs are classified into two categories, namely single-walled nanotubes (SWCNTs), composed of a single-sheet wrapping, and multi-walled nanotubes (MWCNTs), composed of a multi-sheet wrapping. Also, the CNTs electrode's active area depends on the diameter of the nanotubes, and the MWCNTs are generally cheaper than SWCNTs [59].

4.6. Graphene

A graphene is an allotropic form of sp^2 hybridised carbon oriented in a two-dimensional plane according to a hexagonal network. Graphene-based electrodes are robust materials characterised by high conductivity and high specific surface area (> 2000 $m^2.g^{-1}$) [60].

4.7. Graphene Oxide and Reduced Graphene Oxide (GO and RGO)

GO is a chemically modified graphene sheet having a two-dimensional structure, a large specific surface area along with excellent mechanical properties and conductivity [61]. While RGO is a form of GO that has been subjected to chemical, thermal or other treatments for reducing its oxygen content, thus leading to RGO which has outstanding electronic properties such as abnormal quantum Hall effect along with astonishing bearer mobility under relatively high concentrations of charge bearer at ambient temperature [56].

5. Employed modifiers of carbon electrodes for the TC detection

During the electrode modification process, the nature of the employed modifier plays a key role in improving the sensitivity, stability, and reliability of the developed sensor. Therefore, several modifiers are designed to fulfill this purpose and are highlighted below.

5.1. Biological modifiers

The modification of carbon-based electrodes upon employing biological compounds leads to the so-called 'biosensor device' which is an analytical tool designed for transforming a bio(electro)chemical phenomenon into a measurable electrical output signal related to the presence of the target analyte to be detected [61].

The aptamer is a short synthetic sequence of oligonucleotides (15 to 60 bases) manifesting in DNA, RNA, nucleotide, or enzyme capable of undergoing complex folding [4,62]. These sequences are flexible and chemically stable, thus permitting an easy linkage and strong interactions are taking place with the target molecules to be detected such as drugs, proteins, peptides, viruses, or bacteria. The aptamers production is typically accomplished through an in vitro process known as 'The Systematic Evolution of Ligands by Exponential Enrichment (SELEX)' [63,64], which is an economical process that does not require any physiological conditions for handling. Compared to other biological, aptamers can be especially effective against specific targets such as TC and OTC thanks to their broad affinity, wide stability, and large specificity.

Taking these advantages into consideration, the modification of carbon electrodes by aptamers as biological sensing platforms provides an efficient method to recognise and detect tetracycline antibiotics. Consequently, the scientific research has been recently oriented towards exploiting aptamers in crude or mixed form as carbon electrodes modifiers, so for developing label-free biosensors able to sensitively detect TCs and OTCs molecules, therefore, Table 1 illustrates a variety of biosensors applied to detect TCs drugs.

Table 1. Biosensors for TC and OTC detection.

Target	Electrode material	Modifier	Transduction technique	Linear range (M)	Detection limit (M)	Ref
TC	Glassy carbon (GC)	*Anti-TC aptamer	CV	4.4×10^{-5} to 4.44×10^{-8}	2.25×10^{-9}	[66]
TC	Screen printed carbon (SPC)	76-mer 5′-amino-modified oligonucleotide	CV	4.5×10^{-8} to 1.13×10^{-13}	7.86×10^{-11}	[68]
OTC	GC	Anti-TC aptamer immobilized by diazonium-coupling	DPV	2.17×10^{-4} to 4.6×10^{-10}	5×10^{-10}	[75]
TC	GC	Guanine (DNA)	DPV	9×10^{-5} to 3.0×10^{-7}	2.7×10^{-7}	[62]
TC	Graphie paste (GP)	Anti-TC aptamer /oleic acid (OA)	EIS	1×10^{-7} to 1×10^{-12}	3.0×10^{-13}	[30]
			DPV	1×10^{-7} to 1×10^{-10}	2.9×10^{-11}	
OTC	GP /Multi Walled Carbon Nano Tubes (MWCNTs)	Horseradish peroxidase (HRP) (**enzyme**)	DPV	$1.5\times10^{-3}$to 1.5×10^{-7}	3.5×10^{-8}	[81]
TC	GC	Anti-TC aptamer/ MWCNTs-COOH	DPV	5×10^{-5} to 1×10^{-8}	5×10^{-9}	[93]
TC	GC	Anti-TC aptamer/ poly L-glutamic acid /MWCNTs	DPV	1×10^{-6} to 1×10^{-16}	3.7×10^{-17}	[98]
TC	GC	Anti-TC aptamer/ glutaraldehyde/ {Chitosan-prussian blue-graphene}$_2$/MWCNTs-chitosan	DPV	1×10^{-3} to 1×10^{-10}	0.56×10^{-11}	[102]
TC	GC	Anti-TC aptamer/ Graphene oxide	EIS	1×10^{-5} to 1×10^{-13}	2.9×10^{-14}	[103]
OTC	GC	AuNPs/GR-3D/ 5′-SH-CGA CGC ACA GTC GCT GGT GCG TAC CTG GTT GCC GTT GTG T/HRP)	DPV	4.34×10^{-6} to 1.09×10^{-12}	1.08×10^{-12}	[28]
TC	GC	Avidin-HRP / biotin-DNA / thiolated anti-TC aptamer/ MoS_2-TiO_2@AuNPs	DPV	6.0×10^{-6} to 1.5×10^{-10}	5×10^{-11}	[108]
TC	GC	Anti-TC aptamer/ AuNPs /Methylene Blue	CV	1×10^{-3} to 1×10^{-10}	4.2×10^{-12}	[110]
TC	GC	Molecular imprinted polymers/ thiolated aptamer / AuNPs	EIS	1×10^{-6} to 1×10^{-9} and 1×10^{-10} to 0.5×10^{-12}	1.44×10^{-13}	[112]
OTC	GC	HRP-AuNPs -GO- PANI	DPV	2.17×10^{-12} to 8.7×10^{-12}	5.64×10^{-12}	[113]
TC	Pencil graphite	Thiolated aptamer/ AuNPs/Reduced GO	EIS	1×10^{-6} to 1×10^{-16}	3×10^{-17}	[114]
TC	Magnetic bar/GP	Anti-TC aptamer/Fe_3O_4 NPs/activited oleic acid (OA)	EIS	1×10^{-6} to 1×10^{-14}	3.8×10^{-15}	[30]
			DPV	1×10^{-6} to 1×10^{-12}	3.1×10^{-13}	
TC	SPC	Anti-TC aptamer immobilized in RGO/ Fe_3O_4 alginate film	DPV	5×10^{-6} to 1×10^{-9}	0.6×10^{-9}	[161]

Anti-TC aptamer: 5′-GGA ATT CGC TAG CAC GTT GAC GCT GGT GCC CGG TTG TGG TGC GAG TGT TGT GTG GAT CCG AGC TCC ACG TG-3′ (5′-NH_2 modified).

* ***Anti-penicillin aptamer***: 5′-NH_2 GGG AGG ACG AAG CGG AAC GAG ATG TAG ATG AGG CTC GAT CCG AAT GCG TGA CGT CTA TCG GAA TAC TCG TTT TTA CGC CTC AGA AGA CAC GCC CGA CA-3′.

5.1.1. Direct immobilisation of aptamers on the carbon electrode surface

Zhang et *al*. [65] detected the TC molecule utilising a label-free electrochemical aptasensor fabricated upon the deposition of anti-TC aptamer on the surface of a glassy carbon electrode (GCE) [40,66], to ensure the binding of this aptamer on the GCE surface, the activation of its amino groups was established by a carbodiimide coupling reaction with 1-ethyl-3-(3-dimethyl aminopropyl) carbodiimide (EDC) and homobifunctional N-hydroxy succinimide esters (NHS) [67], this reaction results in the formation of phosphate amino ester of the oligonucleotide phosphate group which facilitates the immobilisation of the aptamer on the GCE surface, the anti-TC aptamer has a specific affinity to tetracycline, which induced, in the case of the TC presence in the cell-matrix, an electrochemical signal reaching a detection limit up to 2.25×10^{-9} M, according to the authors, the detection of TC by the prepared electrochemical aptasensor is a sensitive, efficient, easy-to-use and, overall, an inexpensive method. Another label-free electrochemical aptasensor was constructed by Alawad et *al* [68]. for the reagentless detection of TC, the aptasensor is based on the surface modification of a screen-printed carbon electrode (SPCE) by a 76-mer 5′ amino-modified oligonucleotide, the immobilisation process on the SPCE surface was carried out through a cross-linking reaction of carbodiimide to activate the carboxylic groups and producing an amide linkage [69]. The binding between TC and the 76-mer 5′ amino-modified oligonucleotide was studied by cyclic voltammetry (CV), where a rise of the TC oxidation peak is gotten reaching a detection limit up to 7.86×10^{-11} M.

The thin layer of DNA was also employed as an electrode modifier for designing a label-free electrochemical biosensor intended for TC detection, owing to its strong ability to interact with TC, DNA is a relevant targeting system dedicated to the recognition of this molecule in complex media, leading directly to a sensitive detection [70–73]. Nevertheless, DNA possesses a poor mechanical resistance as well as being hydrosoluble which makes its fixation in the aqueous medium a difficult task. Therefore, finding a suitable method to make the DNA hydrophobic is needed.

Through UV irradiation (at 245 nm), Gholivand and Khani proposed a novel method for cross-linking a guanine film extracted from native calf thymus, utilising CV and DPV techniques, they investigated the modification effect of the irradiated film on the CGE surface aimed at the TC detection, the sensor reached a detection limit of 2.7×10^{-7} M, this finding was related to the electrostatic interaction between the irradiated guanine film and the TC molecule, which gives the GCE surface a negative charge, resulting in a huge charge transfer at the interface, thereby improving the sensitivity and selectivity of the prepared sensor [4].

Another label-free aptasensor aimed at the electrochemical detection of TC was fabricated by Jahanbani and Benvidi [5], they immobilised an oligonucleotide on the sensor platform via a free carboxyl group belonging to a double layer of oleic acid (OA), the latter is part of the fatty acids family widely renowned for having a large hydrocarbon tail filled with acidic functional groups [74]. Thus, the combination of OA and graphite paste (GP) surface leads to better deposition of anti-TC-aptamer which increased the sensor's sensitivity towards the TC detection, the EIS and DPV investigations of the anti-TC aptamer/OA/GP were successful and showed detection limits of 3.0×10^{-13} M and 2.9×10^{-11} M, respectively.

For oxytetracycline (OTC) detection, a cost-effective and simple label-free approach for inventing an electrochemical aptasensor was proposed by Y. Yang et *al*. [75]. The

constructed aptasensor is based on modified GCE of which the surface was covered by a layer of anti-TC aptamer grafted via a covalent binding during a diazonium-coupling reaction, so to generate suitable functional groups for OTC binding, the aptamer's immobilisation via diazonium cross-linking reaction is advantageous because of its simplicity and quickness, along with forming resistant and stable aptamer layers against electrical potentials as well as high temperatures [76–78], which provides preciseness and applicability of the sensor, the latter led to a sensitive recognition platform of which the OTC detection limit is 5×10^{-10} M.

The HRP, an electrocatalytic enzyme [79,80] has been used for constructing a label-free electrochemical-biosensor targeting the OTC detection in the work of Ghodsi et *al*. [81], they homogenised the HRP enzyme with a GP/MWCNTs based electrode, where the operating detection concept is based on oxidating the OTC phenolic functional group via the HRP electrocatalytic activity with the presence of H_2O_2. Hence, the resulting oxidation currents, transduced by differential pulse voltammetry (DPV), confirmed the sensitivity and efficiency of the proposed method reaching an OTC detection limit of 3.5×10^{-8} M.

5.1.2. Immobilisation of aptamers along with CNTs on the carbon electrode surface

The presence of nanomaterials as MWCNTs in a label-free electrochemical biosensor has a key role for the biomolecular fixation and the electron mediation [82] providing a high electron transfer and an increase of the faradaic current [83] between the electrode surface and the electroactive analyte upon binding, leading to an improvement of the detection accuracy. Hence, such improvement is linked to the high conductivity, the extremely fibrous structure, and the chemical stability of MWCNTs [84–86]. So, the application of these nanomaterials allowed the development of electrochemical biosensors [83,87–89], i.e. aptasensors [90] and enzyme-sensors [86] for detecting and quantifying TCs in complex matrices.

The electrode surface modification upon functionalising MWCNTs is a way to increase the sensitivity of electrochemical aptasensors applied for the detection of TCs. Thus, MWCNTs functionalised with carboxyl groups are greatly appreciated for this interest [91,92] due to their affinity towards aptamers leading to strong binding of a said biomolecule on the electrode surface, and so enhancing the faradaic current proportional to the TC amount. Zhou et *al*. [93] detected the TC molecule employing a biosensor manufactured by the deposition of anti-TC aptamer on the surface of a GCE covered with carboxyl functionalised MWCNTs, the sensor reached a detection limit of 5×10^{-9} M, the aptasensor was constructed upon ultrasound dispersion of MWCNTs in nitric and sulphuric acids, followed by denaturation of the anti-TC aptamer and passivation with ethanolamine to immobilise the aptamer on the GCE and to eliminate the non-specific adsorbed analytes that can interfere in the detection process, the electrochemical investigation of the label-free anti-TC aptamer/MWCNTs-CO_2H/GCE reveals several advantages, viz. direct response relative to the TC presence without undergoing multiple labelling or washing steps and high chemical stability allowing the reusability of the aptasensor.

The improvement of the aptasensor's electrocatalytic properties depends on the strong fixation of the aptamer. The anti-TC aptamers immobilisation by the biopolymerization process of the MWCNTs surface has known huge consideration [38,94] owing to its rapidity and biocompatibility. Poly Glutamic Acid (PGA) [95] is a hydrophilic biopolymer

composed of monomers linked together via α-amino and free δ-carboxylic acids which attract the anti-TC aptamers and enhances their attachment to the sensor surface [38,96,97]. Benvidi et *al* [98]. developed a highly sensitive electrochemical aptasensor for the determination of TC (LOD of 3.7×10^{-17} M). Employing CV, the manufactured aptasensor allowed a fast one-step electrochemical deposition of PGA on the MWCNTs/GCE surface for immobilising the anti-TC aptamer, the polymerisation of MWCNTs/GCE surface with a biocompatible material such as PGA provides appropriate functional groups to attach the anti-TC aptamer to the electrode surface improving the detection limit.

Recently, the synthesis of nanocomposites for the detection signal amplification have known a wide application for fabricating electrochemical biosensor platforms [99]: ***Chitosan (CS)***, a linear polysaccharide composed of several chains of amino groups that can be easily linked to MWCNTs for forming biocompatible nanotubes [100], ***Prussian blue (PB)***, a hexacyanoferrate compound widely used in electrochemistry because of its typical electrical and magnetic properties [101], and ***graphene (GR)***, a one-thick-atom of bonded carbon that presents wonderful chemical, electrical and biocompatibility characteristics, are suitable materials for forming biocompatible nanocomposites with a high film-forming ability that supply an abundant interface of amino functions necessary for aptamer immobilisation on the electrode surface. Guo et *al*. [87] developed a label-free electrochemical aptasensor based on 5′-amine aptamer immobilised on nanobiocomposite MWCNTs-CS-PB-GR modified GCE, the synergy of these materials has facilitated the electron transfer on the electrode interface leading to an increase in the TC sensing parameters: 0.56×10^{-11} M as detection limit.

5.1.3. Immobilisation of aptamers along with GO nanosheets on the carbon electrode surface

Given their relevant role, GO nanosheets consist of valuable needed materials to enhance the sensing characteristics. Benvidi et *al* [102]. have implemented a label-free anti-TC impedimetric aptasensor for TC detection, the sensor is based on anchoring the GO nanosheets to a GCE surface via diazonium functionalization through epoxy ring-opening, electrostatic attraction, or hydrogen bonding, then, the functionalised GO nanosheets facilitated the aptamer immobilisation on the GCE due to π-π interactions between their hexagonal cells and the aptamers rings [61,103,104], the sensor's detection limit was of 3×10^{-17} M.

5.1.4. Immobilisation of aptamers along with gold nanoparticles (AuNPs) on the carbon electrode surface

As previously mentioned, the use of nanomaterials for biosensor construction is extremely beneficial owing to the excellent properties that they can provide to the electrode interface, which boosts the sensitivity. Thanks to their biocompatibility, large specific surface, and excellent conductivity (the basic driving force that facilitates the oligonucleotides immobilisation with an improvement of the electron transfer on the biosensor surface), AuNPs are highly employed in biosensing applications [105]. Despite the above-mentioned advantages, some studies have found that it's needed to combine AuNPs with other materials to improve the binding affinities to TC molecules [106].

Liu et *al*. [3] have constructed an OTC electrochemical biosensor based on AuNPs/ three-dimensional graphene (GR-3D) film modified GCE. An aptamer, namely: 5′-SH-CGA CGC ACA GTC GCT GGT GCG TAC CTG GTT GCC GTT GTG T-horseradish peroxidase (OTC antibody-AuNPs-HRP) was immobilised on the film. The CV and EIS studies affirmed that the composite synergy constitutes an electrochemical probe with high affinity and ultra-sensitivity towards OTC, promoting higher electron transfer and charge capacity, consequently increasing the analytical performances of the biosensor, reaching a detection limit of 1.08×10^{-12} M.

Molybdenum disulphide (MoS_2) and titanium dioxide (TiO_2) were also amalgamed with AuNPs for biomolecules sensing, i.e. Tang et *al*. [107] created a suitable TC electrochemical biosensor reaching a LOD of 5 x 10^{-11} M, the recognition system consists of a GCE modified upon anchoring via gold-sulphur bonds, avidin-HRP, biotin-DNA and thiolated anti-TC aptamer on ternary nano-composite containing MoS_2-TiO_2 and AuNPs, the investigation showed that the hybridisation of the specific oligonucleotides and ternary nano-composite induces preservation of biomolecule bioactivity as well as an increase of the charge amounts, encouraging the linking to TC and reducing interference-effect.

For maximising the aptamer immobilisation by creating more adsorption sites on the GCE surface, methylene blue (MB) was embedded in AuNPs film covered with anti-TC aptamer via Au-S bonds [108], Yemin et *al*. [109] have adopted this approach as a recognition tool to develop a label-free electrochemical aptasensor for fast and effective TC detection reaching a LOD of 0.42×10^{-11} M.

Molecularly imprinted polymers (MIP) are polymers that undergone an imprinting process to create cavities that have a typical affinity towards a selected target analyte [110], i.e. Rad and Azadbakht [111] have constructed a TC impedimetric aptasensor based on a GCE modified by a nanohybrid amalgam containing a protective layer of MIP, AuNPs, and thiolated aptamer, the synergy of these materials entices a high sensitive platform presenting a low detection limit of 1.44×10^{-13} M under optimal conditions.

Polyaniline (PANI), a conductive polymer exploited for protecting the receptors' surface and improving the electrical conductivity of a prepared biosensor, was utilised as a nanocomposite along with graphene oxide (GO) covered on AuNPs surface for the HRP lock-in, this process was used to prepare a highly sensitive electrochemical OTC biosensor, Xu et *al*. [112] immobilised the GO-PANI mixture on a GCE via electrostatic adsorption forces while the AuNPs deposition was performed via an electrochemical stripping at a constant potential (E = – 0.2 V). The CV studies revealed a higher current density on the HRP-AuNPs-GO- PANI modified electrode compared to the bare one. The increase in the biosensor performances is linked to the interactions between the AuNPs and amino groups of PANI (Au-NH_2) creating OTC binding sites. The sensor reached a LOD of 5.64×10^{-12} M.

Aiming at the direct detection of TC, an impedimetric label-free aptasensor based on modified pencil graphite electrode (PGE) was manufactured by Razdari et *al*. [113], the electrode modification was carried out by depositing AuNPs/Reduced graphene oxide (RGO)/ thiolated aptamer nanocomposite on the PGE. The charge transfer resistance evolution relative to the biosensor during TC detection revealed a fascinating augmentation in the case of the modified electrode compared to the bare one, which is due to the binding between TC and the aptamer improving the electrocatalytic performances and leading to a stable and selective detection of TC with a LOD of 3 x 10^{-17} M.

5.1.5. Immobilisation of aptamers along with magnetite nanoparticles (Fe_3O_4NPs) on the carbon electrode surface

Due to their high specific surface and magnetic dipole interaction, magnetic materials have known huge exploitation by electrochemists [114,115]. Magnetic iron oxide a.k.a. magnetite nanoparticles (Fe_3O_4NPs) are one type of the magnetic materials. However, because of the lack of functional groups, viz. carboxylic, amines, and esters in these nanoparticles [116], the latter are not able to ensure a proper binding with aptamers during sensor assembly, which results in low sensitivity. Indeed, combining magnetite nanoparticles with other materials rich in functional groups required to maintain a congenial environment for binding with aptamers turned out to be compulsory.

Yan et *al*. [117] showed the surface functionalization feasibility of magnetite nanoparticles through double-layer deposition of oleic acid, following this prospect, Jahanbani and Benvidi [5] created a TC electrochemical aptasensor based on activated nanocomposite film of magnetite and oleic acid modified GPE, the electrode contained a magnetic bar to establish a magnetic field. The film activation was performed to ensure an outstanding surface functionality, which subsequently strengthened the anti-TC aptamer crosslinking on the GPE surface. The activation of oleic acid is performed via the carbodiimide coupling reaction through the action of N-Hydroxysuccinimide (NHS) associated with 1-Ethyl-3-(3-dimethylamino propyl) carbodiimide (EDC) which is already reported in this review [67], hence, via EIS and DPV transduction, LODs of 3.8×10^{-15} M and 3.1×10^{-13} M, respectively were gotten.

Xuejia et *al*. have improved the linking between Fe_3O_4NPs and anti-TC aptamer, the method involved incorporating Fe_3O_4NPs into RGO sheets, followed by overlaying the sheets with a film of sodium alginate gel, a natural and biocompatible polymer. To build a TC label-free electrochemical aptasensor, the above-prepared film was implemented on a screen-printed carbon electrode (SPCE) so to immobilise the anti-TC aptamer by physical adsorption. Hence, under optimal conditions, the interaction between the aptamer and TC induced a highly sensitive detection with LOD up to 0.6×10^{-9} M [118–120].

5.2. Chemical modifiers

Chemical modification of carbon-based electrodes is a way for designing sensitive recognition systems intended for the electrochemical detection of TC and OTC. Hence, many chemical compounds as nano-carbon materials, clays, or else polymers have been employed for this purpose. Tables 2 and 3 illustrates therefore a variety of chemically modified carbon electrodes based electrochemical sensors applied for the TC and OTC sensitive detection.

5.2.1. Modification with carbon nanomaterials

For detection performance enhancement, the employment of carbon nanomaterials, whether raw or functionalised as modifiers for fabricating electrochemical sensors intended for TC recognition is beneficial. To optimise the carbon nanomaterials characteristics, i.e. conductivity, catalytic activity, and specific surface area, many studies were carried out regarding the possibility of combination with other conducting nanoparticles, viz. metallic nanoparticles, oxides, and ionic liquids [121–125].

Table 2. Chemosensors for TC and OTC detection.

Target	Electrode material	Modifier	Transduction technique	Linear range (M)	Detection limit (M)	Ref
TC	GC	PtNPs supported on carbon	CV	4.4×10^{-5} to 9.99×10^{-6}	4.28×10^{-6}	[126]
TC	GC	AuNPs/MWCNTs	DPV	1.40×10^{-4} to 5×10^{-6}	4.2×10^{-8}	[127]
TC	GC	CNTs-CO_2H/ZnO	ECL	1.0×10^{-4} to 1.0×10^{-8}	6.7×10^{-9}	[139]
TC	GP	MWCNTs functionalized with CO_2H /GO	adsDPV	3.1×10^{-4} to 2.0×10^{-5}	3.6×10^{-7}	[140]
OTC	GC	MWCNTs/[BMIM]BF_4/ NPs	adsSWV	6.0×10^{-6} to 0.9×10^{-6}	$1.5x10^{-7}$ and $2.0x10^{-8}$	[134]
TC	GC	MWCNTs/ 1–octyl–3–methylimidazolium–hexafluorophosphate	LSV	2.2×10^{-5} to 1.1×10^{-7}	3×10^{-8}	[133]
TC	GC	Potato starch/ carbon black	DPV	120×10^{-6} to 5.0×10^{-6}	1.15×10^{-6}	[151]
OTC	GP	Acetylene black NPs/ montmorillonite with cetyltrimethyl-ammonium bromide in the electrolyte	DPV	50×10^{-6} to 0.5×10^{-6}	8.7×10^{-8}	[143]
OTC	GP	ZnNPs/ montmorillonite	DPV	4.0×10^{-5} to 8.0×10^{-7}	1.2×10^{-7}	[147]
TC	GC	Fe-ZnNPs/ montmorillonite	DPV	5.2×10^{-5} to 3.0×10^{-7}	1.0×10^{-7}	[148]
TC	GC	Polymelamine/RGO	DPV	2.25×10^{-4} to 5.0×10^{-6}	5×10^{-6}	[154]
TC	GP	Polyurethane	DPV	40×10^{-6} to 4.0×10^{-6}	2.8×10^{-7}	[157]
TC	GP	Polyurethane	DPV	3.80×10^{-6} to 95.0×10^{-6}	2.6×10^{-6}	[158]
OTC				3.80×10^{-6} to 95.0×10^{-6}	1.7×10^{-6}	

Table 3. Real-life TC and OTC bio(chemo)sensors.

Target	Electrode material	Modifier	Transduction technique	Application	Recovery %	Reference
TC	GC	Anti-TC aptamer	CV	Milk	1 ng/ml (LOD)	[66]
OTC	GP	HRP/MWCNTs	DPV	Pharmaceutical tablet	106.2-96.6	[81]
				Cow serum	106.1-98.33	
OTC	GC	5′ -NH_2 modified aptamer	DPV	Milk	87.0-99.3	[75]
TC	GC	Fe/Zn–MNT clay	DPV	Fish	103	[148]
				Shrimp	97.6	
				Chicken	99.0	
				Chicken feed	97.2	
TC	GC	ZnO/CNT-CO_2H	ECL	Fishpond water	97.2 - 97.6	[139]
TC	GC	Avidin-HRP / biotin-DNA oligonucleotide / thiolated anti-TC aptamer/ MoS_2-TiO_2@Au	EIS	Milk	90 - 97	[108]
OTC	GP	MNT/Cetyltrim-ethylammonim bromide/ AB	DPV	Fish	105.10	[143]
				Shrimp	105.10	
				Chicken	103.20	
				Chicken feed	98.90	
TC	SPC	Anti-penicillin aptamer/anti-TC aptamer	CV	Natural water	87.8 - 97	[68]
TC	GC	PMA/ RGO	DPV	Human urine	——	[154]
TC	GC	PtNPs	DPV	Human urine	95	[126]
OTC	GC	Anti-TC aptamer/ act / 4-carboxyphenyl diazonium salt	DPV	Whole Milk	87.0- 98.9	[75]
				Skimmed Milk	89.1- 99.3	
TC	GC	Anti-TC aptamer*/ poly L-glutamic acid /MWCNTs	DPV	Honey	95-94	[98]
TC	GC	Anti-TC aptamer*/ GO	EIS	Blood serum	95-97	[103]
TC	GC	Guanine (DNA)	DPV	Milk	98.3-101.0	[62]
				Pharmaceutical sample	99.2-101.0	
TC	GP	PU	DPV	Human urine	98 ± 2	[158]
OTC				Bovine urine	97 ± 2	
				Human urine	96 ± 2	
TC	GP	PU	DPV	Natural water	95	[157]
TC	GP	GO/MWCNTs-CO_2H	DPV	River water	94.0 - 98.5	[140]
				Artificial urine	92.0 - 96.2	
				pharmaceutical formulations	93.4 - 106.0	
TC	GC	Anti-TC aptamer/ AuNPs /MB	CV	Milk	94 - 108	[110]
OTC	GC	MWCNTs /[BMIM]BF_4/AuNPs	Ads SWV	Trout salmonid muscle	5.3x10^{-10} (LOD)	[134]
TC	GC	MWCNTs/ IL	LSV	TC tablet	92.6 - 102	[133]
				Eggs	91.0 - 105	
TC	Magnetic bar/GP	Anti-TC aptamer/Fe_3O_4 NPs/ activited OA	EIS	Serum solution	97 - 104	[30]
				Honey	95 - 96	
				Milk	97 - 103	

** Anti-penicillin aptamer: 5′-NH_2 GGG AGG ACG AAG CGG AAC GAG ATG TAG ATG AGG CTC GAT CCG AAT GCG TGA CGT CTA TCG GAA TAC TCG TTT TTA CGC CTC AGA AGA CAC GCC CGA CA-3′.*

** Anti TC aptamer: 5′-NH_2-AAC GTT TCA GCG CAG GCG CTC ACG GTT GCG GTT GGT CCG CGT GGA TCC GAG CTC CAC GTC CGC ACG GCC GCA CTG C-3′.*

**IL: 1–octyl–3–methylimidazolium–hexafluorophosphate*

5.2.1.1. Combined carbon nanoparticles and metallic nanoparticles based TC sensors. Kushikawa et *al* [126]. have modified a GCE by supported platinum nanoparticles on carbon (PtNPs/C) to develop a TC electrochemical sensor, the characterisation of both the bare and modified electrodes was carried out through CV and DPV, where a high TC oxidation current was observed on the PtNPs/C/GCE meaning that strong reactivity associated to high electron transfer took place on the electrode interface due to the conductivity of PtNPs/C, it has been confirmed that embedded PtNPs/C on the GCE has improved the sensitivity and facilitated the TC diffusion, resulting in high sensing accuracy along with a LOD of 4.28×10^{-6} M. In a more or less similar work, Palisoc et *al* [127]. have built a TC electrochemical sensor based on modifying a GCE through AuNPs supported on MWCNTs, due to the synergic action [127,128] of AuNPs conductivity that increases the charge transfer on the interface, and the adsorption capacity of MWCNTs that traps TC, a significant improvement in terms of TC sensing on the AuNPs/MWCNTs/GCE was observed via DPV technique leading to a LOD of 4.2×10^{-8} M.

5.2.1.2. Combined carbon nanomaterials and ionic liquids based TC and OTC sensors. The combination of carbon nanomaterials and ionic liquids (ILs) is a way of getting relevant TC detection platforms since ILs improve the target analyte's adsorption, promote charge transport at the electrode/solution interface, and inrease the sensitivity [42]. ILs are long chains of alkyl groups easily adsorbed on an electrode surface through hydrophobic electrostatic attraction. ILs are also known for their high conductivity, negligible vapour pressure, thermal, and chemical stability making their use beneficial in the field of TC sensors [129–131].

Gaiping et *al*. [132] developed a TC voltammetric sensor based on modifying GCE by a film of MWCNTs and IL, namely: 1–octyl–3–methylimidazolium–hexafluorophosphate. The synergic effect of MWCNTs and IL facilitates the TC oxidation at 0.54 V in an adsorption controlled irreversible system, increasing the background current density and the sensor sensitivity. Under optimal conditions, employing the LSV technique, a linear relationship between the anodic peak and TC amounts was observed, reaching a LOD of 3×10^{-8} M.

Edgar et *al*. [133] constructed an OTC electrochemical sensor based on modified GCE through depositing MWCNTs filled with AuNPs soaked up with various ILs, namely: 1-ethyl-3-methylimidazolium trifluoromethanesulfonate ($[EMIM]F_3MSO_3$), 1-butyl-3-methylimidazolium tetrafluoroborate ($[BMIM]BF_4$), 1-butyl-3-methylimidazolium hexafluorophosphate ($[BMIM]PF_6$), and 1-butyl-3-methylimidazolium trispentafluoroethyltrifluorophosphate ($[BMIM]3F_5EPF_3$). By using the ads SWV technique, the OTC oxidation signal increased depending on the charge of the used ILs: $3F_5EPF_3^- < F_3MSO_3^- < PF_6^- < BF_4^-$. The highest OTC detection performance was gotten for the $[BMIM]BF_4$/MWCNTs/GCE, manifesting in two linear ranges with LODs of 1.5×10^{-7} M and 2×10^{-8} M.

5.2.1.3. Combined carboxylated carbon nanomaterials and oxide nanoparticles based TC sensors. As reported above [93], carboxylated carbon nanotubes (CNT-CO_2 Hs) are a functionalised version of nanocarbons subjected to a chemical treatment process [134]. Indeed, the combination of these nanoparticles with oxides like zinc oxide (ZnO) or GO is an asset to enhance the performance of an electrochemical sensor [135–137].

Based on CNT-CO_2Hs combined with a flower-like ZnO modified GCE, a TC solid-state eco-friendly ECL sensor was designed by Shan et *al*. [138]. Taking into account the ECL operational principle already described in this review, ZnO herein served as a luminophore and CNT-CO_2Hs as a catalyser to upgrade the ECL intensity, while adding TC to the electrolyte, a decline in the ECL intensity of CNT-CO_2Hs/ZnO/GCE was observed, confirming the deposition of a significant TC amount on the sensor surface owing to the large specific surface area of CNT-CO_2Hs/ZnO composite, consequently heightening the sensitivity and decreasing the interference-effect. The sensor reached a LOD of 6.7×10^{-9} M within a fast response time of 60 s.

Wong et *al* [139]. have exploited the synergic effect of cylindrical MWCNT-CO_2Hs combined with GO thin sheets to design a TC selective electrochemical sensor. The MWCNT-CO_2Hs/GO/GPE revealed a rise in the redox currents intensities due to the presence of large blocks on the electrode surface responsible for TC adsorption employing the ads DPV technique. Thus, improving the sensitivity, selectivity, and stability of the prepared device leading to a detection limit of 3.6×10^{-7} M under the optimal conditions.

5.2.2. Modification with clays

The modification of carbon-based electrodes upon the use of clays for the design of TC-based electrochemical sensors has attracted the researchers' attention, i.e. montmorillonite (MNT), an expansible smectite clay of lamellar structure consisting of an octahedral plate interposed between two tetrahedral silica plates [140]. Thanks to its swelling characteristics, large specific surface area, high cation exchange ability, and strong adsorption capacity in aqueous suspensions [141], MNT is widely employed in the construction of modified carbon-based electrodes, facilitating interactions with the target analyte to be detected and providing binding sites that optimise the TC/OTC detection process.

Sun et al. [142] combined MNT and acetylene black nanoparticles (ABNPs), a subtype of carbon black (CB) renowned for its excellent electric conductivity and high accumulation efficiency [143], the mixture has been employed as a GPE modifier aimed at the fabrication of an OTC electrochemical sensor, the OTC detection in presence of cetyltrimethylammonium bromide was monitored by the DPV technique, displaying a well–defined OTC oxidation peak at 0.6 V along with an enhancement of the background current [144], the latter is explained by the controlled diffusion of one electron and one proton of OTC, along with its adsorption on the porous surface of the modified electrode, the adsorption is controlled via ion-exchange between the positively charged cetyltrimethylammonium bromide on the positive surface of MNT/ABNPs composite and the OTC, leading to its attachment on the sensor surface through hydrophobic C-H bonds, consequently reaching a LOD of 8.7×10^{-8} M.

As discussed, the MNT consists of an octahedral layer of $M(O, OH)_6$ sandwiched between two tetrahedral layers of SiO_4, where M: Al^{3+}, Mg^{2+}, Fe^{2+}, or Fe^{3+}. Generally, due to the isomorphic substitutions between these layers, the failures in binding, and the structural imperfections, negative charges balanced by counter-ions appear at the MNT, which are adsorbed between clay-bearing mineral layers in the infill region. Then, via cationic-exchange reactions, the above cations can be easily substituted by other metal cations to improve the MTN properties [145]. Following this prospect, Sun et al. employed zinc nanoparticles (ZnNPs) to modify MNT via the cation-exchange process, for

constructing an OTC electrochemical sensor based on GCE, which reached a LOD of 1.2×10^{-7} M [146]. Gann et al. adopted the same idea to improve the MNT properties utilising mixed iron and zinc nanoparticles (Fe-ZnNPs), the substituted clay was coated on the GC surface for designing a TC electrochemical sensor, which gave a LOD of 1.0×10^{-7} M [147]. In both works, the modified GCE showed a significant increase of the oxidation current intensity related to OTC or TC, compared to the bare GCE, indicating a catalytic capacity improvement of the modified GCE due to ZnNPs and FeNPs having high specific energy and power density along with high specific surface area [148].

5.2.3. Modification with conductive polymers

Thanks to their unique and advantageous electrical conductivity, conductive polymers have been considerably employed for the TC electrochemical sensors manufacturing over the last decade [149], i.e. potato starch (PS), a biodegradable and biocompatible carbohydrate polymer, was blended with CB to optimise the mechanical and conductive properties for providing a highly electroactive layer, Delgado et al. modified a GCE utilising this layer for developing a TC electrochemical sensor, the surface characterisation of the PS/CB/GCE showed a homogeneous texture and uniform porosity, which are compulsory for the TC adsorption and binding. Indeed, a fast electron transfer has been observed, which improved the detection platform's sensitivity to reach a LOD of 1.15×10^{-6} M for TC [150].

The 1,3,5-triazine-2,4,6-triamine, a.k.a polymelamine (PMA), an organic compound that can be easily electropolymerized and that has the capability of interacting with various target molecules owing to its three amine groups containing nitrogen molecules surrounded by π electrons [151,152]. In the work of Kesavan et al., the PMA was subjected to an agglomeration by RGO layers to produce a stable and conductive film, which was later implemented as a GCE modifier for constructing a TC electrochemical sensor [153], the PMA/RGO/GCE exhibited high selectivity towards TC in the presence of major interferences, along with an increase of the TC oxidation currents and a potential shift towards less positive potential values compared to the bare GCE, the high sensitivity of the sensor was linked to the PMA electropolymerization by the RGO rich in oxygen functional groups that boosted the number of the active sites [154], resulting in a LOD of 5×10^{-6} M.

Polyurethane (PU), a premium quality polymer formed by a repetition of urethane monomers via a simple and cheap reaction of di/polyisocyanate and diol or polyol, PU is generally known for its decent physical and chemical properties [155]. PU was exploited by Calixto et al. as a GPE modifier for designing a TC voltammetric sensor, the DPV signals of the GPE and PU/GPE manifested in two strong peaks at 0.96 V and 1.1 V respectively, explained by the oxidation of the TC phenolic moiety, the PU played an important role in improving the catalytic properties of the GPE, ensuring noble detection of TC with a LOD of 2.8×10^{-7} M [6]. The PU/GPE was reused by the same research group for the detection of TC and OTC, the DPV technique showed strong and separated oxidation peaks of TC at 0.938 V/1.078 V and OTC at 0.943 mV/ 1.078 mV, along with an increase in the background current intensities, reaching LODs of 2.6×10^{-6} M for TC and 1.7×10^{-6} M for OTC, the prepared PU/GPE has a long life-time and does not require surface renewal while performing successive detection tests which makes this sensor very reproducible and valuable for real applications [156].

6. Real-life TC and OTC electrochemical sensors

The sensor's detection capability in real samples is a major tool for validating its analytical reliability [157,158,159,160]. Owing to their simplicity, low-cost, and specificity, the below-discussed sensors are presenting high recoveries (≥ 87%), and are applied for the real determination of TC and OTC employing standard addition methods [93], in fields like medical and therapeutic analysis, environmental monitoring, and food safety.

7. Conclusions and future prospectives

Herein, we overviewed the design of label-free TC/OTC electrochemical biosensors based on carbon electrodes modified by oligonucleotides (aptamers, DNA, or enzyme) carried out by their immobilisation via cross-linking reactions of carbodiimide, diazonium-coupling reactions, UV irradiation, or via free carboxyl groups belonging to OA. Also, the employment of nanomaterials as GO nanosheets, MWCNTs, AuNPs mixed with MB, PANI polymer, RGO, 3D-GR/MoS_2/TiO_2, or Fe_3O_4NPs combined with OA or RGO sheets, showed high electron transfer kinetics and an increase of the faradaic current between the electrode surface and the electroactive analyte (TC or OTC) upon binding leading to an improvement of the detection accuracy.

The chemical modification of carbon-based electrodes is another pathway that has been highlighted for designing TC and OTC recognition systems, the electrode surface modification is usually carried out by the deposition of metallic nanoparticles (AuNPs, PtNPs), carbon nanomaterials mixed with ILs, ZnO, GO, MNT clays, or else polymers. The discussed electrode modifications enhanced the TC/OTC electrochemical detection by assuring high electrical conductivity along with wide specific surface area.

Sensitive, efficient, easy-to-use, and cheap electrochemical bio(chemo)sensors based on modified carbon electrodes designed for TC and OTC detection were deliberated in the present overview, their actual applications in food, pharmaceutical, medical, and environmental samples were discussed, along with obtaining high recoveries confirming their excellent analytical accuracy. However, real challenges are relying on the fabrication of realistic commercial sensor prototypes, that can compete with the conventional detection methods in terms of applicability, sensing characteristics, and low cost.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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